



ELISA Validation Report

Determination of LT3114 in Monkey K₂EDTA Plasma By ELISA

QPS Project Number: 344-1403

Client Project Number: NA

Sponsor

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ELISA VALIDATION REPORT

REPORT TITLE:

Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

QPS PROJECT NUMBER:

344-1403

VALIDATION INITIATED:

9 Jul 2014

CONTRIBUTING SCIENTISTS:

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1. SIGNATURE PAGE

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9/8/14

Date

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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58; the Organization for Economic Co-operation and Development (OECD) Principles of Good Laboratory Practice [C (97) 186/Final]; the Japanese Ministry of Health, Labor, and Welfare (MHLW); Ordinance No. 21, March 26, 1997; and the FDA Guidance for Industry: Bioanalytical Method Validation, Sept 2013.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.

Melissa Calvarese

Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC

9/3/14

Date

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3. QUALITY ASSURANCE STATEMENT

QPS 344-1403

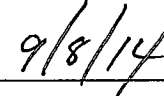
Protocol Title: Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1403 validation protocol and standard operating procedures, and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Draft Protocol	Y Li	2 Jul 2014	2 Jul 2014
Study Records/ Draft Report	R. McCrae	13,14 Aug 2014	14 Aug 2014



Rita McCrae M.S.
Quality Assurance Auditor
QPS, LLC



Date

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4. VALIDATION SUMMARY FOR LT3114

An immunoassay method for the determination of LT3114 in monkey K₂EDTA plasma was developed and validated. Experiments were conducted at QPS to evaluate the precision and accuracy of the assay in monkey plasma.

The ELISA assay utilized a sandwich format, in which, microtiter plates are pre-coated with LPA-BSA Conjugate. The plate is washed. Samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by HRP conjugated goat anti-human IgG monkey adsorbed. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl.

Assay range, intra-batch and inter-batch precision / accuracy data, along with the stability, dilution linearity, and matrix evaluation data are provided in the following summary table. Additional long term frozen stability will be evaluated at a later date, and will be added as an addendum.

Report Title	Determination of LT3114 in Monkey K ₂ EDTA Plasma by ELISA
Report Number	QPS 344-1403
Analyte Name	LT3114
Sample Volume:	5µL (Before 100x dilution, duplicate measurement)
Analytical Method Type	ELISA
Calibration Range:	100 – 2000 ng/mL
QC Concentrations	100, 300, 750, 1500, and 2000 ng/mL
QC Intra-batch Precision (%CV)	1.9% - 5.3%
QC Intra-batch Accuracy (%RE)	-3.0% - 13.0
QC Inter-batch Precision (%CV)	5.4% - 12.9%
QC Inter-batch Accuracy (%RE)	-3.3% - 4.0%
Room temp Stability in Monkey K ₂ EDTA Plasma	24 Hours at Room Temperature
Freeze/thaw Stability in Monkey K ₂ EDTA Plasma	5 Cycles at -70°C and -20°C/ Room Temperature
Long-term Storage Stability in Monkey K ₂ EDTA Plasma	30 Days at -20°C and -70°C
Dilution Linearity	900,000 ng/mL diluted 1, 100, 500, 1000, and 5000 -fold
Selectivity (10 lots, spiked 100, 2000 ng/mL)	100% lots tested within 100±25% Recovery

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5. OBJECTIVE

The purpose of this study was to validate an ELISA to determine the concentrations of LT3114 in monkey K₂EDTA plasma samples collected during a nonclinical study.

6. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

AQL	Above Quantifiable Limit
BQL	Below Quantifiable Limit
Exc	Sample excluded according to QPS Validation Protocol 344-1403
LLOQ	Lower Limit of Quantification
LLOQ QC	Lower Limit of Quantification Quality Control
MRD	Minimum required Dilution
NC	Not Calculated
NA	Not Applicable
NR	Not Reported
OD	Optical Density
QC	Quality Control
r²	Coefficient of Determination
SD	Standard Deviation
SOP	Standard Operating Procedure
ULOQ	Upper Limit of Quantification
ULOQ QC	Upper Limit of Quantification Quality Control

$$\text{Precision} \quad \% \text{Coefficient of Variation (\%CV)} = 100 \times \frac{\text{Standard Deviation}}{\text{Mean Concentration}}$$

$$\text{Accuracy} \quad (\% \text{RE}) = 100 \times \frac{\text{Mean Found Concentration} - \text{Nominal Concentration}}{\text{Nominal Concentration}}$$

Calibration standard and Quality control samples at five concentration levels were prepared in matrix and were determined based upon the actual concentrations of the stock solutions.

All statistics in the data tables are calculated by Watson™ LIMS or based on the “precision as displayed” option of Microsoft® Excel (e.g., Mean, S.D., %CV, %Diff).

7. ANALYTE INFORMATION

Name: LT3114
Concentration: 20.14^a mg/mL
Purity: 96%
Supplied Form Lpath
Lot Number: LP121416
Expiration Date: 4/30/16 (assigned per QPS SOP LP-007)^b
Storage Conditions: -70°C

^aWith purity applied, concentration will be used at 19.33mg/mL.

^bNo supporting documentation available for stability or retest date of compound at current storage condition.

8. ACCEPTANCE CRITERIA

The validation acceptance criteria and the statistical data will be determined at a minimum to QPS' SOPs TLM-001 Validation of Binding Assays and QPS validation protocol 344-1403. Additional experiments (if any) not governed under QPS' SOPs will follow the instructions provided by the sponsor.

Six successful, complete precision and accuracy runs (out of which three are consecutive runs) are required to validate a method. The evaluation of an intra-batch validation run includes a minimum of six replicates of QCs at each concentration and a minimum of five QC concentrations (including LLOQ and ULOQ). The evaluation of inter-batch validation runs (minimum of six runs) includes a minimum of three replicates of QCs at each concentration per run. Each run must include a minimum of five QC concentrations.

Method Acceptance: A method is accepted if it successfully fulfills the following:

- Intra-batch Accuracy and Precision acceptance criteria: For each intra-batch validation run, the mean of six or more replicates at each concentration of the QC samples is not allowed to deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value. In addition, the intra-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Inter-batch Accuracy and Precision acceptance criteria: The mean value at each QC concentration must not exceed $\pm 20.0\%$ ($\pm 25.0\%$ at LLOQ and ULOQ) from its nominal value. In addition, the inter-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Dilution test acceptance criteria: The mean of the back calculated concentration (after the dilution factor has been applied) must be within

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100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered acceptable.

- Assay selectivity acceptance criteria: At least 80% of the lots spiked with LT3114 should have recovery between 75-125%.
- Room temperature and freeze/thaw stability acceptance criteria: The mean back calculated concentration must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.
- Frozen stability samples acceptance criteria: The back calculated concentration is compared to the nominal value of the QC samples, which was confirmed in the initial analysis as properly prepared. The mean value for each concentration level must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.

Run Acceptance criteria:

- For QCs, validation samples and future study samples, each replicate will be measured twice and the average will be reported if the %CV from the two measurements is $\leq 25\%$. Otherwise, the value from the two measurements will not be reported.
- Production runs or sample analysis runs are accepted based on the performance of calibration standards and QCs. Only standards back-calculated to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) different from their nominal concentrations will be excluded from the regression analysis. If more than one standard back calculates to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ), exclusion should be carried out one standard at a time to see if the other standards would fall within $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) of the nominal value after the exclusion.

If both of the calibration standards at the lowest (or highest) concentration of the curve are excluded from the regression analysis due to various reasons (e.g., not enough sensitivity at the lower end, curve bend at the higher end, QCs or standards at the specific concentration fail to meet acceptance criteria), the LLOQ (or ULOQ) shall be moved up (or down) to the next standard concentration in sequence (also known as the truncation of the standard curve).

Finally, a minimum of six standard concentrations must be within the assay range. Excluded calibration standards that fall outside of the calibration range shall not be used to evaluate the acceptability of the run. All excluded

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calibration standards over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) within the calibration range should be reported (e.g., as Exc) and should be included when evaluating the acceptability of the run.

Anchor points may be included or excluded to facilitate curve fitting based on scientific judgment.

At least 2/3 of the back calculated QC concentrations must be within $100 \pm 20\%$ ($100 \pm 25\%$ at the LLOQ and ULOQ QCs) of their nominal value, 50% of QC at each concentration must not deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value.

- Special case of run acceptance for accuracy and precisions run:
The precision and accuracy runs are accepted solely based on the performance of the calibration curve. A minimum of 75% of the total number of calibration standards in the calibration range (between the LLOQ and ULOQ of the run) should be within $100 \pm 20\%$ ($100 \pm 25\%$ at LLOQ and ULOQ standards) of their nominal value, and a minimum of six unique standard concentrations must be within the assay range. The calibration curve must contain at least one calibration standard at both the LLOQ and ULOQ of the range.

9. MATERIALS

9.1. Sample-Handling Equipment

- 96-well microtiter plates (Costar Cat #2592)
- 96-well format micro titer/dilution tubes
- 25-100 mL Reagent reservoir
- Plate sealers
- Polypropylene plastic tubes: various volumes
- Pipette tips for single and multichannel pipettes
- Graduated cylinder: various volumes
- 8-Channel StatMatic II microplate dispenser

9.2. Analytical Equipment

Spectramax Microplate Reader (Molecular Devices).

10. METHODS

10.1. Preparation of Secondary Stock Solutions:

Two sets of stock solutions were prepared separately for standards and QCs. The prepared stock solutions were used immediately to prepare subsequent working standard and QC solutions and the remaining portion was discarded.

10.2. Preparation of Working Standard Solutions in Monkey K₂EDTA Plasma

Appropriate amount of stock was combined with monkey plasma according to the following table to make the calibration standard working solutions. Each concentration of the standards

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contains at least 95% (v/v) monkey plasma. The prepared working standard solutions were aliquoted for single use and were stored at -70°C. The blank pooled monkey plasma served as the zero standard (Blank).

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of MKPL (μL)	Total Volume (μL)
0	0	200	200
50	5	395	400
100	5	195	200
200	10	190	200
400	20	180	200
800	40	160	200
1000	50	150	200
1400	70	130	200
1800	90	110	200
2000	100	100	200
4000	200	0	200

10.3. Preparation of Working QC Solutions in Monkey K₂EDTA Plasma

Appropriate amount of stock was combined with monkey plasma according to the following table to make the QC working solutions. Each concentration of the QCs contains at least 95% (v/v) monkey plasma. The prepared QC solutions were aliquoted for single use and were stored at -70°C.

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of MKPL (μL)	Total Volume (μL)
100	5.0	195.0	200
300	75.0	925.0	1000
750	37.5	162.5	200
1500	375	625.0	1000
2000	100	100.0	200

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10.4. Analytical Method

<u>Parameter</u>	<u>Value of Parameter</u>
General:	
Analyte	LT3114
Matrix	Monkey K ₂ EDTA Plasma
Calibration standard concentrations	50(AP), 100, 200, 400, 800, 1000, 1400, 1800, 2000 and 4000(AP) ng/mL
Quality Control Concentrations	100, 300, 750, 1500, and 2000 ng/mL
Minimum required dilution	100x in Dilipidized Bovine Serum
Sample volume	5 µL (before dilution, 2 wells) 100 µL/well (after dilution)
ELISA procedure summary	Microtiter plates are pre-coated with LPA-BSA. After washing, samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by horseradish peroxidase conjugated goat anti-human IgG H and L, Monkey Adsorbed Ab. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl. The color intensity is proportional to the quantity of LT3114.
Data Acquisition:	
Equipment	Molecular Devices SpectramaxPLUS ³⁸⁴
Program	Softmax Pro Version 3.1.2
Reading mode	Endpoint
Wavelength	450 nm (reading), 630 nm (reference)
Calibration	Auto-calibration on
Blanking	No pre-read, Plate blank not used
Mixing	Off
Analysis and Reporting:	
Program	Watson™ LIMS Version 7.4.1
Regression type	4-parameter (auto-estimate)
Weighting:	1
Anchoring Point	Use as needed.
Precision for Reporting:	3 significant digits

11. RESULTS AND DISCUSSION

Table 1 lists all the validation runs performed.

11.1. Regression Analysis

Calculations were done using 4-parameter (auto-estimate) regression using Watson™ LIMS (v. 7.4.1, Thermo Electron Corporation, USA) Table 2 lists the back-calculated calibration standard concentrations for LT3114 and Table 3 lists the standard curve parameters for

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LT3114. The results show a 4-Parameter fit from 100-2000 ng/mL for LT3114. Figure 1 shows a typical calibration curve of LT3114 in monkey plasma.

11.2. Precision and Accuracy

Precision and accuracy of the method were determined by analyzing QC samples at a minimum of five different concentrations within the standard curve range.

Precision was determined by the coefficient of variation (%CV) and accuracy was determined by percent recovery (%RE). The acceptable maximum %CV for a valid assay is 25% at all concentrations. The acceptable accuracy (%RE) for a valid assay is within $100 \pm 20\%$ (25% at LLOQ and ULOQ QCs) of the nominal concentration.

The intra-batch precision and accuracy within an analytical run were determined by analyzing six replicates for each QC sample including the ULOQ and the LLOQ QC from the same run. The results for intra-batch precision and accuracy (Table 4) were acceptable.

The inter-batch precision and accuracy of the assay over different analytical runs were determined by analyzing QC samples from six validation runs with three replicates in each run. The results for inter-batch precision and accuracy (Table 5A) were acceptable. In addition the results for QCs from all runs are summarized in Table 5B.

11.3. Lower Limit of Quantification

The lower limit of quantification (LLOQ) of the assay was 100 ng/mL for LT3114. For sample analysis, concentrations found below the LLOQ after the minimum required dilution will be reported as 'BQL', if applicable, since the precision and accuracy of the assay were determined to that limit.

11.4. Stability

11.4.1. Room Temperature Stability

Room temperature stability of LT3114 in monkey plasma was performed to evaluate LT3114 stability in monkey plasma for the time that the samples would be at room temperature during sample handling and processing. Stability of the QC samples was determined at two concentrations (low and high QCs) in minimum three replicates at both 6 and 24 hours. The results (Table 6) indicate that LT3114 in monkey plasma is stable at room temperature for at least 24 hours.

11.4.2. Freeze/Thaw Stability

The stability of LT3114 in monkey plasma after five freeze/thaw cycles was determined. QC samples at two concentrations (Low and high QCs) were frozen at -20°C or -70°C (for a minimum of 24 hours for cycle 1 and a minimum of 12 hours for later cycles) and thawed at room temperature. This freeze/thaw cycle was repeated five times. After the completion of the fifth freeze/thaw cycle, the samples were analyzed and compared to the nominal value. During initial testing of the freeze thaw samples the -20°C Low QC recovered lower than expected, the test was repeated at n=6, and was found to recover within the acceptable range.

The results (Table 7A and Table 7B) indicate LT3114 is stable for five freeze/thaw cycles in monkey plasma under the described conditions.

11.5. Long-Term Frozen Storage Stability

LT3114 quality control samples in monkey plasma were stored at -70°C and -20°C for 30 days before being analyzed against a freshly prepared calibration standard curve. The results (Table 8) indicate LT3114 is stable for at least 30 days in monkey plasma at -70°C and -20°C. The frozen storage stability of LT3114 in monkey plasma at -70°C and -20°C at longer periods will be assessed at a later date and an addendum to the report will be provided.

11.6. Dilution Linearity and Hook Effect

The purpose of the dilution test was to ensure that samples could be diluted with monkey plasma without affecting the final calculated concentration. Monkey plasma samples prepared at one concentration 900,000 ng/mL were diluted with monkey plasma at dilution factors of 1, 100, 500, 1000, and 5000 in three replicates at each level. The results were corrected with the dilution factor and compared to the nominal concentration. The results in Table 9 show that the difference between the mean of the adjusted concentration (found concentration multiplied by dilution factor) and the nominal concentration of LT3114 was within the acceptable range. There was no hook effect observed.

11.7. Assay selectivity in Monkey K₂EDTA Plasma

Ten individual lots of monkey plasma were spiked with or without LT3114 and measured for LT3114 concentration. Table 10 indicates that 100% of the tested lots had a recovery between 75-125%, which is acceptable.

12. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

DEV-14-376: In step 8 “Plate Reading” the Analytical Procedure specified Lm2 as 570nm, however the plates were all read using 630nm as Lm 2. The method development for this project was consistent with the 630nm wavelength used during validation. Throughout the study analyst and PI incorrectly initial dated that the 570nm was used as the Lm2. There is no impact on study data and a note to the record was added the study notebook for clarification.

13. CONCLUSION

An ELISA method (QPS method 344-1403) has been validated for the quantitation of LT3114 in monkey K₂EDTA plasma from 100 – 2000 ng/mL. With a validated dilution factor of 5000, this assay is suitable for measuring LT3114 in monkey K₂EDTA plasma from 100 – 10000000 ng/mL.

14. REFERENCES

Method Validation Protocol:

QPS Project No. 344-1403

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SOPs:

QPS LP-004: “Analytical Standards and Quality Controls”

QPS LP-007: “Analytical Reference Compounds”

QPS LP-034: “Receipt and Handling of Biological Matrices”

QPS TLM-001: “Validation Quantitative of Ligand Binding Assays”

Notebooks:

QPS validation notebook 344-1403

LT3114 assay notebook

15. ARCHIVE STATEMENT

All raw data (including electronic SoftmaxPro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark DE, 19711, USA. Study records will be archived according to the most current version of SOP GN-014, “Archive”

16. TABLES

Table 1 Validation Run Log

Curve Number	Analysis Date	Run Description	Accepted/Rejected ^a
1	9-Jul-14	140709-3441403-MKPL-MAC-1A-001	Accepted
2	9-Jul-14	140709-3441403-MKPL-MAC-2B-002	Accepted
3	9-Jul-14	140709-3441403-MKPL-GP-1A-003	Accepted
4	9-Jul-14	140709-3441403-MKPL-GP-2B-004	Accepted
5	10-Jul-14	140710-3441403-MKPL-EEQ-1A-005	Accepted
6	10-Jul-14	140710-3441403-MKPL-EEQ-2B-006	Accepted
7	11-Jul-14	140711-3441403-MKPL-MAC-1A-007	Accepted
8	11-Jul-14	140711-3441403-MKPL-MAC-2B-008	Accepted
9	16-Jul-14	140716-3441403-MKPL-MAC-1A-009	Accepted
10	17-Jul-14	140717-3441403-MKPL-MAC-1A-010	Accepted
11	7-Aug-14	VOID ^b	NA
12	7-Aug-14	140807-3441403-MKPL-MAC-1A-012	Accepted

a: Acceptance based on QPS Validation Protocol 344-1403

b: Run voided due to planning error, processed as run 012

**Table 2 Back-Calculated Calibration Standard Concentrations of LT3114
Monkey K₂EDTA Plasma**

Analysis Date	Curve Number	Concentrations (ng/mL)							
		100	200	400	800	1000	1400	1800	2000
9-Jul-14	1	105	204	387	788	968	1380	1760	2060
		100	205	385	774	1090	1450	1760	1970
9-Jul-14	2	98.7	197	383	798	1010	1420	1780	2010
		98.6	202	392	800	997	1430	1780	1980
9-Jul-14	3	94.0	190	392	831	1090	1430	1730	1980
		96.2	185	376	814	1050	1460	1700	1950
9-Jul-14	4	98.8	206	396	810	978	1410	1800	1990
		96.8	203	385	801	1040	1390	1790	2000
10-Jul-14	5	107	204	407	806	1010	1410	1780	2030
		Exc	199	400	778	998	1400	1780	1990
10-Jul-14	6	97.6	211	409	792	1010	1390	1800	2030
		99.8	203	404	796	1000	1390	1780	2020
11-Jul-14	7	99.4	200	408	794	1010	1400	1850	1990
		100	191	405	779	1020	1360	1830	1960
11-Jul-14	8	104	197	402	800	1040	1410	1830	1980
		98.0	192	386	791	991	1400	1780	2000
16-Jul-14	9	96.8	221	423	787	970	1420	1820	2020
		87.7	212	421	787	967	1440	1750	2020
17-Jul-14	10	103	192	416	895	931	1400	1770	2090
		110	177	425	777	968	1370	1810	2000
7-Aug-14	12	97.9	201	408	789	1010	1400	1770	2040
		96.8	205	397	802	1010	1360	1820	2000
n		21	22	22	22	22	22	22	22
Mean		99.3	200	400	800	1010	1410	1790	2010
S.D.		4.70	9.58	13.8	25.1	38.5	26.5	34.6	32.6
%CV		4.7	4.8	3.5	3.1	3.8	1.9	1.9	1.6
%RE		-0.7	0.0	0.0	0.0	1.0	0.7	-0.6	0.5

Exc: Excluded according to QPS Validation Protocol 344-1403

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Table 3 **LT3114 Standard Curve Regression Parameters**

Analysis Date	Curve Number	Min	Max	Slope	Ed50	R-Squared	LLOQ	ULOQ	Regression Footnote(s)
9-Jul-14	1	0.077196810	9.200081202	1.073971400	10571.368443463	0.9991	100	2000	1
9-Jul-14	2	0.063511273	38.477247641	0.935907175	68685.749423884	0.9998	100	2000	1
9-Jul-14	3	0.064120884	4879.830532427	0.993463228	9874195.481421820	0.9985	100	2000	1
9-Jul-14	4	0.059013501	34184.680661393	1.010770010	60369913.375127300	0.9997	100	2000	1
10-Jul-14	5	0.071299796	15.840082125	1.068198645	23422.851722878	0.9997	100	2000	1
10-Jul-14	6	0.086739283	14.801481479	1.073822412	23017.011802598	0.9999	100	2000	1
11-Jul-14	7	0.079744960	15.359883908	0.999031598	21081.313630949	0.9996	100	2000	1
11-Jul-14	8	0.072687304	14.789715323	0.979564736	19582.881777004	0.9998	100	2000	1
16-Jul-14	9	0.095728082	7.747141688	1.146963989	6431.959550719	0.9996	100	2000	1
17-Jul-14	10	0.340156524	6.546377670	1.143375025	3878.050907112	0.9985	100	2000	1
7-Aug-14	12	0.176919055	7.278381753	1.069744952	5673.028396259	0.9996	100	2000	1

Regression Footnote(s):

1) Resp. = (Min - Max) / (1 + (Conc. / Ed50) ** Slope) + Max

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Table 4 Intra-Batch Precision and Accuracy of LT3114 in Monkey K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
n		6	6	6	6	6
Intrarun Mean		113	335	781	1510	1940
Intrarun SD		3.35	17.7	31.9	28.1	84.5
Intrarun %CV		3.0	5.3	4.1	1.9	4.4
Intrarun %RE		13.0	11.7	4.1	0.7	-3.0

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Table 5A Inter-Batch Precision and Accuracy of LT3114 in Monkey K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
9-Jul-14	2	97.5	267	735	1410	1910
		95.2	269	711	1400	1890
		94.6	278	714	1460	1980
9-Jul-14	3	89.4	270	719	1400	1860
		76.1	245	652	1230	1720
		84.0	252	713	1370	1860
9-Jul-14	4	98.6	383	810	1440	1970
		90.7	327	705	1430	2040
		96.8	293	764	1470	1940
10-Jul-14	5	110	329	843	1580	2140
		96.6	294	765	1380	1920
		NR	309	814	1440	2030
10-Jul-14	6	119	337	851	1550	2140
		106	304	766	1390	1930
		130	389	855	1510	2150
n		20	21	21	21	21
Mean		103	312	767	1450	1960
S.D.		13.3	40.2	54.9	78.6	109
%CV		12.9	12.9	7.2	5.4	5.6
%RE		3.0	4.0	2.3	-3.3	-2.0

NR: Not Repotable %CV>25%

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Table 5B Inter-Batch Precision and Accuracy of LT3114 in Monkey K2EDTA

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
9-Jul-14	2	97.5	267	735	1410	1910
		95.2	269	711	1400	1890
		94.6	278	714	1460	1980
9-Jul-14	3	89.4	270	719	1400	1860
		76.1	245	652	1230	1720
		84.0	252	713	1370	1860
9-Jul-14	4	98.6	383	810	1440	1970
		90.7	327	705	1430	2040
		96.8	293	764	1470	1940
10-Jul-14	5	110	329	843	1580	2140
		96.6	294	765	1380	1920
		NR	309	814	1440	2030
10-Jul-14	6	119	337	851	1550	2140
		106	304	766	1390	1930
		130	389	855	1510	2150
11-Jul-14	7		373	769	1430	
			320	829	1610	
11-Jul-14	8		327	823	1580	
			296	795	1550	
16-Jul-14	9		323	709	1370	
			308	694	1370	
17-Jul-14	10		313	720	1500	
			339	767	1550	
7-Aug-14	12		282	667	1280	
			302	774	1570	
n		20	31	31	31	21
Mean		103	314	763	1460	1960
S.D.		13.3	35.8	54.3	89.7	109
%CV		12.9	11.4	7.1	6.1	5.6
%RE		3.0	4.7	1.7	-2.7	-2.0

NR: Not Repotable %CV>25%

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Table 6 Room Temperature Stability of LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
6 hours	280	1380
	276	1300
	277	1280
n	3	3
Mean	278	1320
SD	2.08	52.9
%CV	0.7	4.0
%RE	-7.3	-12.0
24 hour	268	1180
	263	1280
	253	1230
n	3	3
Mean	261	1230
SD	7.64	50.0
%CV	2.9	4.1
%RE	-13.0	-18.0

Run ID: 140716-3441403-MKPL-MAC-1A-009

Table 7A Freeze/Thaw Stability of LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
-70°C	271	1250
5 Cycles	264	1250
	262	1220
n	3	3
Mean	266	1240
SD	4.73	17.3
%CV	1.8	1.4
%RE	-11.3	-17.3
-20°C	280	1140
5 Cycles	246	1200
	257	1150
n	3	3
Mean	261	1160
SD	17.3	32.1
%CV	6.6	2.8
%RE	-13.0	-22.7

Run ID: 140716-3441403-MKPL-MAC-1A-009

**Table 7B Repeat of -20°C Low QC Freeze/Thaw Stability of LT3114 in
Monkey K₂EDTA Plasma**

<u>Nominal Concentration (ng/mL)</u>	
1500	
Lot: EEQ140708-14	
5 Cycles	1360
	1290
	NR
	1380
	1460
	1390
n	5
Mean	1380
SD	61.1
%CV	4.4
%RE	-8.0

NR: Not Reportable %CV>25%

Run ID: 140717-3441403-MKPL-MAC-1A-010

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Table 8 30 Day Long Term Frozen Stability of LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
-20°C	265	1200
	260	1230
	263	1210
n	3	3
Mean	263	1210
SD	2.52	15.3
%CV	1.0	1.3
%RE	-12.3	-19.3
-70°C	267	1290
	279	1290
	289	1230
n	3	3
Mean	278	1270
SD	11.0	34.6
%CV	4.0	2.7
%RE	-7.3	-15.3

Run ID: 140807-3441403-MKPL-MAC-1A-012

Table 9 Dilution Test of LT3114 in Monkey K2EDTA Plasma

Nominal Concentration (ng/mL)	Dilution Factor	LT3114 (ng/mL)		
900000	1	AQL		
		AQL		
		AQL		
		n	3	
		Mean	NC	
		SD	NC	
		%CV	NC	
		%RE	NC	
		100	AQL	
			AQL	
			AQL	
			n	3
			Mean	NC
			SD	NC
			%CV	NC
	%RE		NC	
	500		1010000	
			1000000	
		1010000		
		n	3	
		Mean	1010000	
		SD	5770	
		%CV	0.6	
		%RE	12.2	
		1000	984000	
			1050000	
	1040000			
	n		3	
	Mean		1020000	
	SD		35600	
	%CV		3.5	
	%RE		13.3	
	5000		1070000	
			973000	
		991000		
		n	3	
		Mean	1010000	
		SD	51600	
		%CV	5.1	
		%RE	12.2	
Cumulative n		9		
Cumulative Mean		1010000		
Cumulative SD		32500		
Cumulative %CV		3.2		
Cumulative %RE		12.2		

Run ID: 140711-3441403-MKPL-MAC-1A-007

AQL: Above Limit of Quantitation (>2000 ng/mL)

NC: Not calculated

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Table 10 Assay Selectivity of LT3114 in Monkey K₂EDTA Plasma

Individual Monkey Plasma	LT3114 Concentration (ng/mL)				
	Blank (0 ng/mL)	Spike-in LLOQ QC (100 ng/mL)		Spike-in ULOQ QC (2000 ng/mL)	
Lot #	Found	Found	% Recovery of Spiked Lot	Found	% Recovery of Spiked Lot
CYN141483	BQL	115	115.0	2490	124.5
CYN141484	BQL	121	121.0	2330	116.5
CYN141485	BQL	117	117.0	2110	105.5
CYN141486	BQL	120	120.0	2200	110.0
CYN141487	BQL	120	120.0	2160	108.0
CYN141488	BQL	120	120.0	2180	109.0
CYN141489	BQL	123	123.0	2210	110.5
CYN141490	BQL	120	120.0	2210	110.5
CYN141491	BQL	122	122.0	2160	108.0
CYN141492	BQL	115	115.0	2100	105.0
n		10		10	
Mean		119		2220	
SD		2.75		116	
%CV		2.3		5.2	
%RE		19.0		11.0	

Run ID: 140711-3441403-MKPL-MAC-2B-008

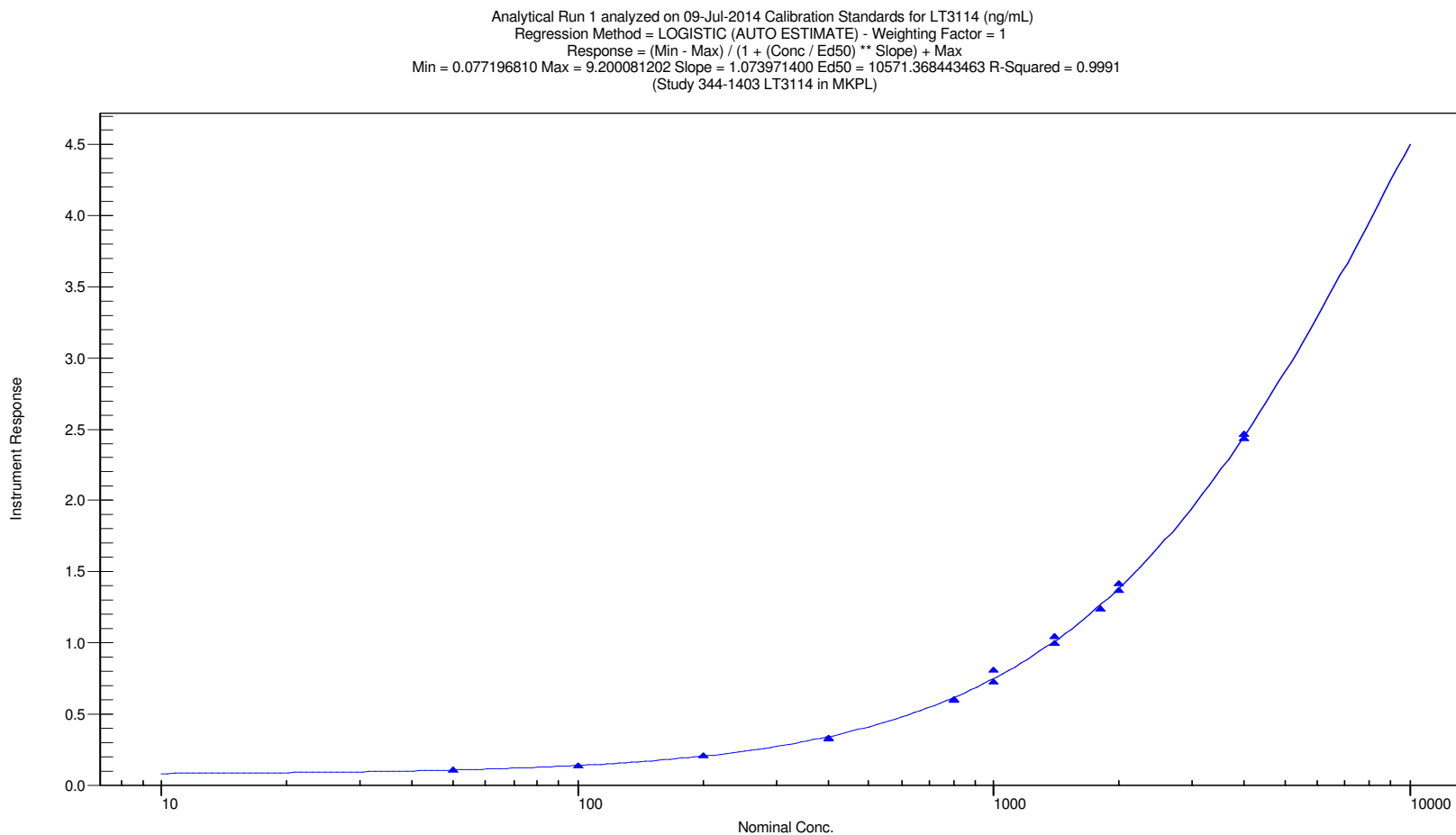
BQL: Below Limit of Quantitation (<100 ng/mL)

% Recovery: (Found Concentration / Nominal of the Spiked) X 100%

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17. FIGURES

Figure 1 **Typical Standard Curve for LT3114 ELISA in Monkey K₂EDTA Plasma**



Run ID:140807-3441403-MKPL-MAC-1A-001

18. APPENDICES

Appendix A Certificate of Analysis of LT3114

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416
Formulation Buffer		Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)		2 to 8 °C
Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

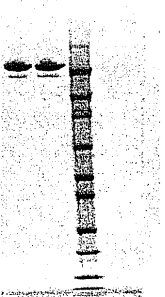
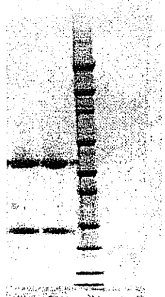
Summary
<u>Production location, date, and method</u> The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C. Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage. Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage. Purified lots LP121416 and LP121417 were pooled to create lot LP121419. <u>Purification location, date and method</u> Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning). <u>Concentration by A₂₈₀ and extinction coefficient</u> The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg. <u>Total amount of Ab (mg)</u> 1400 mg <u>Appearance details:</u> Clear, slightly yellow solution

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Pages 1-5
Rcm 5/2/14

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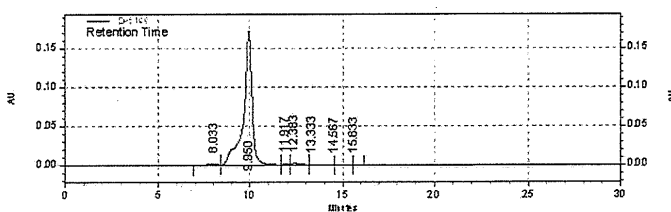
		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Two pronounced bands; One at ~55 kDa; One at ~ 28 kDa
	Insert images of SDS-PAGE	Non-Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty 	Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty 
	Isoelectric Focusing (by IEF Gel)	Report Results	Not done

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		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	3 of 5

Attribute	Test	Specifications	Results																																																			
Potency	Protein Concentration (OD-280)	Report Results	20.14 mg/mL																																																			
	LPA Binding Direct ELISA (EC 50)	Report Results	25.05 ng/mL																																																			
	Bioactivity Assay (IL-6 and EDG2)	Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 µg/mL																																																			
Purity	Insert image of SE-HPLC																																																					
		<table><tr><th colspan="5">Det166 Results</th></tr><tr><th>Time</th><th>Area</th><th>Area %</th><th>Height</th><th>Height %</th></tr><tr><td>8.033</td><td>68097</td><td>1.14</td><td>1413</td><td>0.78</td></tr><tr><td>9.950</td><td>5707101</td><td>95.95</td><td>173377</td><td>96.24</td></tr><tr><td>11.917</td><td>35431</td><td>0.60</td><td>1744</td><td>0.97</td></tr><tr><td>12.383</td><td>97022</td><td>1.63</td><td>2645</td><td>1.47</td></tr><tr><td>13.333</td><td>31081</td><td>0.52</td><td>668</td><td>0.37</td></tr><tr><td>14.567</td><td>8116</td><td>0.14</td><td>234</td><td>0.13</td></tr><tr><td>15.633</td><td>1015</td><td>0.02</td><td>62</td><td>0.03</td></tr><tr><td colspan="2">Totals</td><td>5947863</td><td>100.00</td><td>180143</td><td>100.00</td></tr></table>		Det166 Results					Time	Area	Area %	Height	Height %	8.033	68097	1.14	1413	0.78	9.950	5707101	95.95	173377	96.24	11.917	35431	0.60	1744	0.97	12.383	97022	1.63	2645	1.47	13.333	31081	0.52	668	0.37	14.567	8116	0.14	234	0.13	15.633	1015	0.02	62	0.03	Totals		5947863	100.00	180143	100.00
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15.633	1015	0.02	62	0.03																																																		
Totals		5947863	100.00	180143	100.00																																																	
SE-HPLC	Purity [% as IgG monomer]	≥ 90%	96%																																																			
	Aggregates [as % of total peak area]	≤ 6%	1%																																																			
	Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%																																																			

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		Certificate of Analysis Form			
TITLE Certificate of Analysis for Monoclonal Antibody Produced by Lpath					LOT NUMBER LP121416
DOCUMENT NUMBER FM-COA-13	REVISION 0.0.1	SUPERCEDED REVISION 0.0.0	ISSUE DATE OCT 31, 2013	EFFECTIVE DATE Nov 1, 2013	PAGE 4 of 5

Attribute	Test		Specifications	Results
	SDS-PAGE reduced	Purity [%]	≥ 90%	≥ 90%
		50 kDa Band	~65%	~67%
		25 kDa Band	~35%	~33%
		Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin		≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden		≤ 10 CFU/mL	Not done

Comments																																
Source Data: <table border="1" data-bbox="573 1117 1073 1375"> <thead> <tr> <th></th> <th>Lpath notebook</th> <th>Pages</th> </tr> </thead> <tbody> <tr> <td>Production</td> <td>277</td> <td>149</td> </tr> <tr> <td>Purification</td> <td>296</td> <td>11,18,25</td> </tr> <tr> <td>UV A280</td> <td>296</td> <td>48</td> </tr> <tr> <td>SDS-PAGE</td> <td>296</td> <td>26</td> </tr> <tr> <td>LPA binding ELISA</td> <td>221</td> <td>21-22</td> </tr> <tr> <td>IL-6</td> <td>237</td> <td>48-49</td> </tr> <tr> <td>EDG2</td> <td>237</td> <td>51</td> </tr> <tr> <td>SE-HPLC</td> <td>296</td> <td>33-34</td> </tr> <tr> <td>Endotoxin</td> <td>298</td> <td>13</td> </tr> </tbody> </table>				Lpath notebook	Pages	Production	277	149	Purification	296	11,18,25	UV A280	296	48	SDS-PAGE	296	26	LPA binding ELISA	221	21-22	IL-6	237	48-49	EDG2	237	51	SE-HPLC	296	33-34	Endotoxin	298	13
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SE-HPLC	296	33-34																														
Endotoxin	298	13																														
Additional Comments on Sample Quality:																																

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Certificate of Analysis Form

TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	5 of 5

JUDGMENT

Approved for testing in humans

□ - Accepted

□ - Rejected

**For Research Only
Not For Use in Humans**

Prepared by: 
Cindy Dickerson, Ph.D.

Date: 12-6 2013

Reviewed by: Jon Wojciak, Ph.D.

Date: 12/6/ 2013

Approved by: Gary Woodnutt
Gary Woodnutt, Ph.D.

Date: 3rd Dec 2013

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ELISA Validation Report

Determination of LT3114 in Monkey K₂EDTA Plasma By ELISA

ADDENDUM #1

QPS Project Number: 344-1403

Client Project Number: NA

Sponsor

Lpath Inc.

4025 Sorrento Valley Blvd.

San Diego, CA 92121

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QPS, LLC
Delaware Technology Park
3 Innovation Way, Suite 240
Newark, DE 19711
WEB: www.qps-usa.com
TEL: (302) 369-5601
FAX: (302) 369-5602

ELISA VALIDATION REPORT ADDENDUM #1

REPORT TITLE:

Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

QPS PROJECT NUMBER:

344-1403

ORIGINAL REPORT DATE:

8 Sept 2014

ADDENDUM 1 DATE:

27 Oct 2014

CONTRIBUTING SCIENTISTS:

Melissa Calvarese, B.S., Chad Asher, B.S.

TESTING FACILITY:

QPS, LLC
Delaware Technology Park
1 Innovation Way, Suite 200
Newark, DE 19711

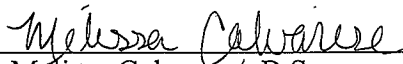
SPONSOR:

Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121

Confidential

1. SIGNATURE PAGE

WRITTEN BY:



Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC

10/27/14

Date

APPROVED BY:



Chad Asher, B.S.
Group Leader and Head of PK Bioanalysis
QPS, LLC

10/27/14

Date

TABLE OF CONTENTS

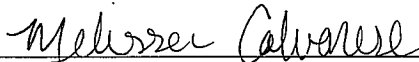
1. SIGNATURE PAGE	3
2. COMPLIANCE STATEMENT	5
3. QUALITY ASSURANCE STATEMENT	6
4. RESULTS AND DISCUSSION	7
Long-Term Stability (LTS)	7
5. TABLES	8
Table 1A -20°C Long Term Frozen Stability for LT3114 in Monkey K ₂ EDTA Plasma	9
Table 1B -70°C Long Term Frozen Stability for LT3114 in Monkey K ₂ EDTA Plasma	10

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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58;; and the FDA *Guidance for Industry: Bioanalytical Method Validation*, May 2001.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.



Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC

10/27/14
Date

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3. QUALITY ASSURANCE STATEMENT

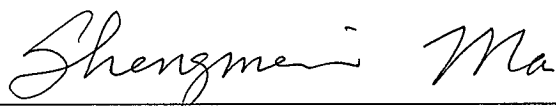
QPS 344-1403

Protocol Title: Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

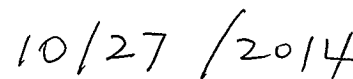
This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1403 validation protocol and standard operating procedures, and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Study Records/Draft Report Addendum 1	S.Ma	15 Oct 2014	15 Oct 2015

The undersigned has assumed that the QPS final report accurately reflects the source data.



Shengmei Ma
Quality Assurance
QPS, LLC



Date

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4. RESULTS AND DISCUSSION

Long-Term Stability (LTS)

LT3114 quality control samples prepared in Monkey K₂EDTA Plasma were stored at -20°C and -70°C for 92 days before being analyzed against a freshly prepared calibration standard curve. The results (Table 1A and 1B) indicate LT3114 is stable for at least 92 days in Monkey K₂EDTA Plasma at -20°C and -70°C.

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5. TABLES

Table 1A

-20°C Long Term Frozen Stability for LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
92 Days	242	1370
	262	1290
	273	1290
N	3	3
Mean	259	1320
SD	15.7	46.2
%CV	6.1	3.5
%RE	-13.7	-12.0

Initial run of QCs: 140709-3441403-MKPL-MAC-1A-001,
to demonstrate proper preparation of LTS samples.

Run ID: 141008-3441403-MKPL-MAC-1A-013

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Table 1B**-70°C Long Term Frozen Stability for LT3114 in Monkey K₂EDTA Plasma**

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
92 Days	263	1240
	243	1270
	251	1310
N	3	3
Mean	252	1270
SD	10.1	35.1
%CV	4.0	2.8
%RE	-16.0	-15.3

Initial run of QCs: 140709-3441403-MKPL-MAC-1A-001,
to demonstrate proper preparation of LTS samples.

Run ID: 141008-3441403-MKPL-MAC-1A-013

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